

# Technical Appendix: An Empirical Analysis of Transfer for Dynamic Path Planning

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## A. Protocols by Stage

All continuous stages share the substrate: a 2-D point robot, maps with procedurally generated obstacles, a probabilistic roadmap (PRM) of exactly 192 nodes with  $k=7$  connectivity, and  $A^*$  evaluated at a per-suite *binding budget* of node expansions calibrated (on the anchor baseline only) so that baseline success falls in the pre-specified discriminative band. Unless noted, training labels are exact graph-Dijkstra costs on the PRM (static) or exact backward space-time Dijkstra values over (node,  $t$ ) (dynamic). Headline continuous runs are local single-GPU (RTX 5090) validations with one primary training seed unless stated; the discrete program ran on a remote fleet.

## A2. Exact Environment, Model, and Data Specifications

**World generation.** Continuous worlds are unit squares (side length 1.0; the large-room family uses 2.0); units are abstract. Start and goal are sampled with minimum separation 0.45 of the side length and inserted as roadmap nodes 0 and 1. Roadmaps contain exactly 192 sampled nodes with  $k=7$  nearest-neighbor connectivity; edges are validated by collision sampling at resolution 0.010 of the side length, and worlds whose roadmap fails to connect start to goal are rejected and resampled.

**Dynamic data collection.** A candidate training world is *usable* iff (i) the world builds, (ii) its roadmap connects start to goal, and (iii) the space-time problem is solvable from the start at  $t=0$  (finite backward space-time Dijkstra value). Training worlds are drawn only from the dynamic maze, rooms, and spiral families (dense maze, crossing, and large rooms are family-level held out from dynamic training). The heavy run draws seeds until 53 usable worlds are collected; each world contributes its reachable ( $v, t$ ) states (192 nodes across  $t_{\max}+1$  time steps, masked to reachable), averaging 21,312 states per world for 1,129,536 supervised scalar samples in total. All three training seeds use the same collection procedure with independent seeds.

## B. Full Static and Dynamic Suite Tables

Tables 6 and 10 give the complete per-suite C7 static and C8 dynamic results summarized in the main paper; Tables 7–9 give the fixed-provider primary view and its replications.

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**Multi-seed replication detail (Table 9).** Two full-pipeline retrains (collect + train, seeds 2001/2002) of the aware/blind field U-Net twins were run under a design frozen before training, then evaluated on the same frozen fresh cohort with the canonical binding budgets. Matched-ratio medians across seeds: maze 0.047–0.059, rooms 0.117–0.135, spiral 0.065–0.092 (seed-stable); crossing 0.273–0.434, dense maze 0.216–0.625, large rooms 0.250–0.548 (seed-variable). Preregistered readouts: R1 (success CI excludes zero in  $\geq 5/6$  suites) 5/6 and 6/6—pass; R2 (deltas within  $\pm 0.15$  of canonical in  $\geq 5$  suites) 5/6 and 6/6—pass; R3 (no suite significantly aware-over-blind) fails as stated for seed 2001 (large rooms +0.080 [+0.020, +0.160])—the same suite that is significantly *blind*-better under seed 1234 (−0.200 [−0.320, −0.100]). Across the 18 seed-suite twin contrasts: 2 significantly aware, 2 significantly blind, 14 null, with the significant cells in different suites for different seeds; the per-suite future-window effects flip sign across seeds and are consistent with training-seed noise. No retraining, reselection, or recalibration occurred in response to any readout.

**Success as a function of budget (Figure 3).** A single evaluation pass at  $4\times$  the binding budget on the same frozen cohort yields *exact* success-vs-budget curves for every lower budget: the space-time  $A^*$  expansion order is budget-independent, and a map is solved at budget  $B$  iff its recorded solve expansions are  $\leq B$ , so the curves are derived by thresholding with no re-runs and no interpolation (design frozen before launch). As a determinism cross-check, thresholding at the binding budget reproduces every fresh-cohort success value exactly (all suites, all providers). The headline result is not an artifact of the calibrated operating point: Euclidean search needs  $1.7\text{--}2.8\times$  the binding budget to reach the fixed blind provider’s binding-budget success (crossing  $2.49\times$ , maze  $2.29\times$ , dense maze  $1.95\times$ , rooms  $1.73\times$ , large rooms  $1.94\times$ , spiral  $2.80\times$ ), and by  $4\times$  the binding budget every provider converges to the same feasibility ceiling (0.94–1.00; on dense maze the 0.94 ceiling binds even the oracle, reflecting maps unsolvable within the time horizon).

## C. Transfer and Composition Details

Representative C9 HRM curves (ratio at success): maze-dense LoRA 0.650 (1.00) at  $K=1$  to 0.730 (0.97) at  $K=16$ ; maze-dense full FT 0.744 (0.76) to 0.571 (0.99) at  $K=16$  and 0.552 at  $K=32$ ; scratch 1.008 (0.37) to 0.808 (0.80);

|                                              |                                                |                                                    |                                              |                                              |
|----------------------------------------------|------------------------------------------------|----------------------------------------------------|----------------------------------------------|----------------------------------------------|
| <b>Representation &amp; training</b>         | C5: scalar HRM collapses to residual cap       | C6: field target rescues HRM (succ. .625→.975)     | C7: well-trained scalar is also viable       |                                              |
| <b>Planner integration</b>                   | Discrete: additive residual inert vs Manhattan | Same weights, focal rank $w=1.0$ : −6–15% exp.     | C7–C8: additive beats focal vs loose Euclid  | C13: fresh certifier 0/6 → shared queue 6/6  |
| <b>Formulation, information, supervision</b> | C8/C9b: future window no benefit (0/9)         | C9/C9h: LoRA plateau = capacity; labels set regime | C11/C12: no depth or hierarchy dose-response | C13-M: bounded obs. + local Bellman: −15.95% |

Figure 1: Program map: the three harness ingredients the study isolates—representation and training, planner integration, and formulation/information/supervision—with the stages that intervened at each and their verdicts (orange: failure or negative; blue: positive under the corrected harness; gray: mechanism or null result).

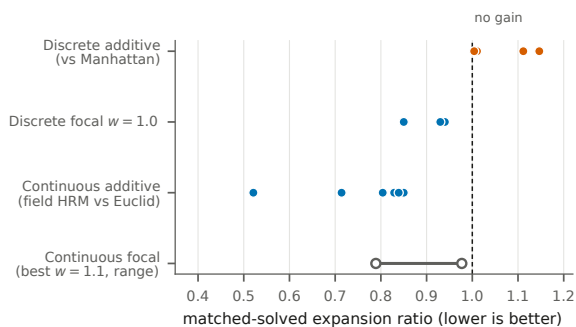


Figure 2: Integration summary across substrates (moved from the main paper). Discrete grids (Manhattan anchor): additive learned magnitude is at or below baseline while the same trained predictions reduce expansions by 6–15% as a within-band ranker at  $w=1.0$ . Continuous roadmaps (Euclidean anchor): additive guidance reduces matched expansions by 15–48%; the plotted focal  $w=1.1$  range is an exploratory, post-hoc operating point and is not used for any main-paper claim.

rooms-large full FT 1.128 (0.37) at  $K=1$  to 0.500 (0.92) at  $K=8$ ; bugtrap LoRA degrades from  $\approx 0.696$  (0.83) to 0.950 (0.48) at  $K=32$ —LoRA is usually but not universally base-preserving. C9h (matched compute): bounded-minus-unbounded LoRA median ratio delta  $0.000 \pm 0.008$  across 27 cells (22/27 exactly zero; max  $|\Delta|$  0.028); rooms-large field U-Net full FT reaches 0.404 at 0.97 success from 0.982 zero-shot while conv-LoRA stays at 0.992–1.002. C9b (dynamics): maze-dense aware U-Net zero-shot 0.145 (0.60), LoRA  $K=16$  0.059 (0.97), full FT  $K=1$  0.112 (0.70); aware-minus-blind success deltas at full-FT  $K=16$  are 0.00 or  $-0.05$  in all nine cells. C10: RBF-merge minus zero-shot expansion-ratio deltas by target/backbone:  $-0.022$ ,  $-0.012$ ,  $-0.024$ ,  $+0.082$ ,  $+0.012$ ,  $+0.116$ ; weight-space vs. prediction-space deltas lie within  $[-0.012, +0.007]$ ; own-axis RBF mass 0.986–0.998.

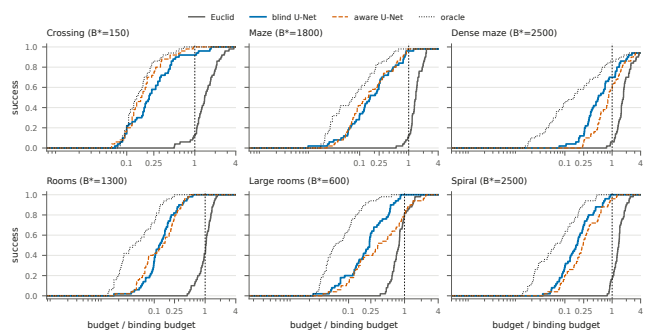


Figure 3: Success-vs-budget step curves per dynamic suite (fresh 50-map cohort), derived exactly from one  $4 \times$ -binding evaluation pass. Budgets are normalized by the binding budget  $B^*$  (dashed vertical line). The learned providers reach high success at  $0.05$ – $0.3 \times B^*$ ; Euclid requires  $1.7$ – $2.8 \times B^*$  to match the blind provider’s binding-budget success.

## D. C11 Gates, Collapse Forensics, and Scale

**Collapse forensics.** Five of 33 recurrent/ACT arm-cells show constant-prediction or partial seed collapse (HRM-trace A8/C8, ON-LSTM B0, HRM-v2 B0, two ON-LSTM A8 seeds), with a reproducible loss signature at the normalization cap. The largest recurrent deficits (HRM-trace ratios 1.526 at A8, 1.600 at C8) are collapse measurements, not expressiveness estimates. Removing  $K=0$  padding *worsens* recurrent predictions: models used pad steps as extra recurrent compute.

**Scaled addendum.** A 12-run `unet_film_big` vs. `hrm_trace_big` comparison (complete; 2,700 eval rows) preserves the ordering: completion 0.804 vs. 0.736 at  $K=2$  and 0.413 vs. 0.307 at  $K=8$ ; mean state MAE 0.395 vs. 0.447 ( $K=2$ ) and 0.906 vs. 1.855 ( $K=8$ ). At  $K=8$  the scaled HRM exactly matches the leg-sum baseline’s completion and mean expansions (0.307; 701.34)—persistent collapse, not recovered hierarchy.

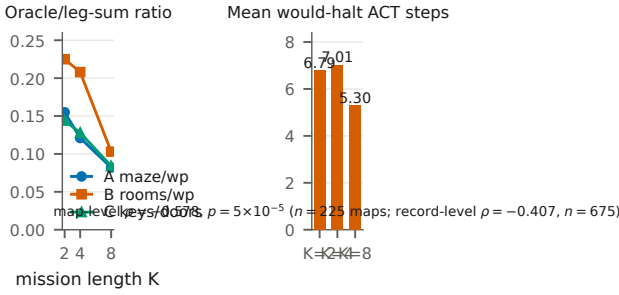


Figure 4: Compositional-mission hierarchy test (C11);  $K$  here is mission length. Left: the exact oracle’s ratio over a leg-sum baseline falls as missions deepen (headroom exists). Right: learned HRM halting uses *fewer* steps at  $K=8$ ; the annotation reports both analysis grains (map-level  $\rho = -0.578$ ,  $p = 5 \times 10^{-5}$ ,  $n=225$  maps; record-level  $\rho = -0.407$ ,  $n=675$ ).

### E. C12 Detail

C12-A (memory headroom, then learned pilot): constructed decision-relevant aliasing at 71.3% of paired decisions; privileged mode/history diagnostic +0.467 completion and  $-65.1\%$  collision-adjusted regret (map-bootstrap 95% CI 63.0–66.9%); oracle completion 0.975. The one-seed matched temporal-hierarchy pilot fails G1 (forecast), G2 (planning), and G3 (persistent carry); overall *strong negative*. C12-B (tied refiner): cycle-1 $\rightarrow$ 8 normalized burden improves monotonically in every cell (A/K2 0.859 $\rightarrow$ 0.815; A/K8 0.624 $\rightarrow$ 0.576; C/K2 0.867 $\rightarrow$ 0.777; C/K8 0.576 $\rightarrow$ 0.517; all cycle-1/8 map-bootstrap intervals separate), but the  $K8-K2$  gain is +0.0038 (CI  $[-0.0199, 0.0281]$ ) on A and  $-0.0305$  ( $[-0.0726, 0.0096]$ ) on C—no dose-response. Tied beats shallow and untied on C/K8 (deltas 0.0177/0.0076, BH  $q=6.7 \times 10^{-5}$ ) but loses to untied on A/K8 ( $q=0.0168$ ). Bellman residual *worsens* over cycles in every cell while burden improves; solved cycle-8 paths average 1.4–2.2% above oracle cost.  $K=16$  was dropped before training by a construction gate (only 2/300 and 1/300 valid missions).

### F. C13: Mechanism Ladder and Frozen Confirmation

**Claim supported.** A provider restricted to bounded local observations (radius-0.20 subgraph on a known roadmap; behavior-derived labels; no shortest-path supervision) can, with one inference-time local Bellman backup, beat every fixed complete-map C7 provider in pooled expansions at matched empirical path quality on a frozen, disjoint cohort.

**Ladder.** Each rung changes one factor; deltas are paired expansions versus the matched comparator (rungs C–G versus matched focal search at  $w=1.10$ , including oracle providers on six development maps; rungs I–M versus the complete-map field HRM, live on all six suites).

**Frozen confirmation, per suite.** 144 maps (24 per family; seeds disjoint from every prior cohort; feature caches hash-verified against live regeneration; model, radius 0.20,

$\alpha=1.50$ , integration, and analysis frozen before generation).

**Scale calibration.** On the disjoint 48-map calibration block,  $\alpha \in \{1.00, 1.25, 1.50, 2.00\}$  gives deltas  $-3.79$   $[-9.40, +1.73]$ ,  $-9.77$   $[-15.38, -4.42]$ ,  $-13.04$   $[-18.67, -7.65]$ , and  $-16.94$   $[-22.56, -11.54]$ , with 4/4/6/6 negative suite means and mean/max cost ratios rising from 1.022/1.241 to 1.052/1.366;  $\alpha=1.50$  was frozen as the largest scale whose suite breadth was complete before the absolute-ceiling verdict, and every arm formally fails the preregistered absolute 1.10 maximum-cost ceiling—the confirmation gate is therefore comparator-relative (mean within +0.005, max within +0.02 of field HRM), which passes with margins to spare (1.0235/1.1624 vs. 1.0311/1.3346).

**Operating points and timing.** The direct  $\alpha=1.50$  arm is the matched-quality expansion result and carries no formal bound. The separate certified operating point runs the learned rank inside a  $w=1.10$  focal band with a shared-queue consistent-bound incumbent proof and passes all 144 bound and certificate checks. Per-map wall time decomposes as feature construction 5.138 s (prototype Python builder), model inference 0.0008 s, local backup 0.0097 s, and search 0.0002 s, versus 0.371 s complete-map field-HRM inference: the result is search effort and path quality, not latency.

**Architecture substitutions under the identical method** (C13-N/O/P) are summarized in the main paper; their raw artifacts (development grids, verdicts, integrity manifests) ship in the artifact package.

### G. Discrete Program Detail

The discrete program is substantially larger than the canonical subset reported here: fifteen experiment families across eleven surveyed cloud volumes (13,671 parsed result files) and roughly thirty experiment scripts, including a ten-run early scaling lineage of obstacle-trajectory forecasters, diffusion path-generator baselines, a superseded harder-calibration predecessor of the structured protocol below, partial precursors of the controlled transfer run, an empty-model baseline-rerun control (static  $A^*$  re-verified through the learned-model code path), and a budget-invariance analysis establishing that raising search budgets adds work without rescuing success on the hard families. The paragraphs below report the completed canonical results; the full inventory ships with the released artifacts.

**Forecasting.** Matched 100-episode protocol: HRM-3M/10M 71/100 vs. LSTM 67–69/100 (no uncertainty reported; two-episode matched gap). Structured Preset M+ (four 100-episode suites): ON-LSTM-10M 0.3875 mean success vs. HRM-10M 0.2650 and HRM-3M 0.200; ON-LSTM-3M also has the lowest rollout MSE (0.1257/1.1269 one-/five-step vs. 0.3973/2.6438 for HRM-3M).

**Additive transfer (clean v3, 22 suites  $\times$  3 budgets).** Manhattan 0.590 success / 126,707 mean expansions; HRM full-FT 0.585 / 128,013; ON-LSTM full-FT 0.564 / 127,228; HRM LoRA 0.554 / 140,852; ON-LSTM LoRA 0.514 / 145,282. Every learned arm  $\leq$  matched baseline.

**Pooled multitask.** HRM pooled base 0.612 success (+2.11pp) / 122,184 expansions vs. Manhattan 0.591 / 126,630—the strongest completed discrete additive positive;

only 1 of several specialists beats its matched four-suite baseline (+2.0pp; others  $-0.42$  to  $-20.25$ pp).

**Focal ranking (local pilot, 3–8 seeds).** Magnitude is scale-flat despite near-perfect ordering (correlation 0.987–0.994; predicted/true mean 0.73/0.42/0.27/0.19 at 64/128/192/256). At  $w=1.0$ : HRM A128-static ratio 0.85 with success 0.62→0.75; A192 0.94 (0.75→0.75); A128-dynamic 0.93 (0.62→0.62); ON-LSTM A128/A192 0.85 (0.75→0.75). At  $w=1.05$ , A128-dynamic improves further but A192 success regresses 0.75→0.62;  $w=1.0$  is the evidence-safe operating point. Experts exactly match pooled bases in tested cells.

## H. Validity Forensics

**The frozen-halt artifact.** A 27.27M-parameter direct maze solver reported 96.64% token accuracy, 25.40% exact accuracy, and “74.60% halt accuracy” with 16 average adaptive-computation steps. Audit:  $74.60 = 100 - 25.40$  exactly—the halt logit was frozen negative (a constant-false classifier) because the training loop omitted the halt loss and back-propagated only the final segment. After repairing the loss path (verified live: halt loss 0.14→0.06, token accuracy 90–95%), exact accuracy remained 0.000 at the chosen stop point; the repaired run demonstrates mechanism liveness, not task convergence. Bit-exact forward parity with the reference implementation was separately established. We recommend the  $100-x$  pattern as a standard red flag for adaptive-computation metrics.

**The cap-collapse signature.** C5’s scalar HRM collapse sits at a loss value pinned by the residual cap (2.612 in that setup, with constant predictions at the cap). The same class of signature reappears in C11: collapsed cells share an *identical* final loss (1.1397, exactly the constant-cap loss under that normalization), enabling automatic collapse detection instead of misreading collapsed cells as capacity evidence.

**Pad steps as compute.** In C11, removing  $K=0$  padding tokens *degraded* recurrent predictions—models had co-opted padding as additional recurrent iterations. Sequence-model comparisons that alter padding alter compute.

## I. Statistics

Success is compared with paired McNemar tests, BH-corrected within stage families. Expansion effort is compared only on matched solved maps via median ratios with bootstrap percentile CIs (10k–20k resamples); from C12 onward, bootstraps are map-clustered by design. C9–C11 originally pooled repeated evaluations of fixed test maps across adaptation/model seeds; Section J reports the map-clustered reanalysis of every quantity the main paper quotes from those stages; the fixed-provider dynamic reanalysis of Table 7 uses the same map-level grain. Matched-solved ratios condition on joint success and can favor arms that solve only easier maps; we therefore always report success and ratios separately and never combine them into a single score. Negative results are failures to observe an effect under the tested protocol, not equivalence claims; no equivalence tests were run.

## J. Map-Clustered Reanalysis

Units are test maps; adaptation/model seeds are averaged within maps before inference; success deltas use 10,000-resample map bootstraps; expansion ratios are medians of per-map ratios over matched maps at the binding budget in additive mode; the C11 halting test aggregates model seeds within maps and permutes mission length within config (maps are independent across  $K$  cells; 20,000 stratified permutations). Estimator conventions differ slightly from the published record-level curves, so point values differ modestly; the question is verdict-level agreement. A methodological caution surfaced during this work: a first pass that accidentally pooled additive and focal evaluation modes attenuated the apparent C9 transfer effects by roughly half—mode/aggregation mixing alone can manufacture or destroy an effect of the size under study.

**C11 halting.** Record level reproduces the published value ( $\rho = -0.4066$ ,  $n=675$ ); map level strengthens it ( $\rho = -0.5778$ ,  $n=225$ , stratified permutation  $p=5\times 10^{-5}$ ). Map-level means 6.78/7.01/5.30 at  $K=2/4/8$ .

**C11 vs-MLP ratios.** Map-clustered ratio-of-means CIs exclude parity at  $K \in \{0, 2\}$  in config A for both U-Net (0.734/0.846) and GNN (0.822/0.923), and in config B at  $K=0$  (0.904/0.818) plus B/ $K=2$  for the GNN (0.805); config C never separates; every  $K \in \{4, 8\}$  cell includes parity.

**C9h bound-vs-capacity.** 27 cells: median bounded-minus-unbounded map-level ratio delta +0.0000; 15/27 exactly zero; sd 0.0077; max  $|\Delta|$  0.0370—the capacity attribution stands.

**C9b aware—blind (full FT,  $K=16$ ).** Eight of nine point deltas are zero or negative; the single positive (+0.017 [0.000, +0.050]) does not exclude zero; no cell shows a significant aware advantage.

**C10 composition.** Map-paired deltas vs. zero-shot: four cells  $\approx 0$  with CIs crossing zero; the two ON-LSTM rooms cells are significantly *worse* (+0.089 [+0.061, +0.121] and +0.098 [+0.076, +0.123]), matching the published point deltas and upgrading the null to “never better, sometimes significantly worse.”

No preregistered verdict flips under any of these recomputations.

## J1. Tuned Weighted-A\* Control

### J2. Fixed-Model Selection Audit Trail

The primary dynamic claim rests on one fixed learned heuristic; its provenance is as follows. (1) *Candidates*: the original dynamic run trained aware and blind twins of scalar HRM, scalar ON-LSTM, and field U-Net models. (2) *Selection*: the field U-Net blind twin was designated the single primary model for every suite during the fixed-model reanalysis design; the original 20-map evaluation results were known at that time, so the accurate statement is that the model, integration mode (additive), and per-suite binding budgets were **fixed before the 50-map confirmation cohort was generated**, not that selection predates all target outcomes. (3) *Freeze*: the confirmation cohort’s generation seed (999,999) was chosen in advance; the evaluation-seed formula’s within-suite offset bound

guarantees map-disjointness from every earlier cohort, and world fingerprints (start, goal, obstacle coordinates) were checked. (4) *Integrity*: the fixed checkpoint’s SHA-256 is `b8378950545f8abdbf06d59568b5d8ab6069884c1c038d23a41a14b6dc17fb6f`; the binding-budget calibration file is copied verbatim into every evaluation directory. (5) *Seeds*: the two additional training pipelines change only the training seed (2001, 2002); collection, architecture, integration, budgets, and the evaluation cohort are identical.

## K. Reproducibility and Artifact Inventory

From C7 onward every stage freezes a written design (gates, budgets, analysis grain) before execution; artifacts (checkpoints, feature caches, cohort manifests, raw rows, verdicts, reports) are hash-bound in integrity manifests; confirmation cohorts are generated only after freezing model, integration, and evaluation code; failed development gates block confirmation rather than triggering retuning. Duplicate-evaluation determinism is enforced where preregistered (byte-identical traces; identical decision rows). The final-stage pilot on persistent planning-time state ran under a fully test-verified harness (158 focused-plus-neighboring tests; hardware preflight). Its first staged execution ended at a preregistered execution-integrity hard stop—the harness’s independent reanalysis disagreed with the primary comparison computation, voiding all mechanism fields. After repair of the named defect only, a fresh-fingerprint rerun passed every integrity primitive, including independent reconstruction of the comparison evidence from promoted raw artifacts, and returned the valid negative reported in the main paper; persistent carry loses to the same-checkpoint reset twin on map-macro frontier ranking (MRR delta  $-0.029$ , 95% CI  $[-0.060, -0.003]$ ), and no self-bootstrap or confirmation was run. We note the initial stop as the integrity machinery functioning as designed.

The anonymized artifact archive submitted with this paper contains: (i) the analysis scripts that regenerate every main-paper figure and quoted map-level statistic from raw rows (the fixed-provider dynamic reanalysis, the map-clustered reanalysis, and all figure generators); (ii) the raw paired evaluation rows behind every quoted quantity (C7/C8 evaluation tables, including the fresh-cohort and three-seed replication rows, C9/C9H/C9B/C10 evaluation tables, C11 evaluation and halting tables, and C13 development and confirmation rows); (iii) map/cohort manifests, seed lists, frozen configurations, and the integrity manifests for the frozen confirmations; (iv) the preregistration design documents from C7 onward; and (v) a top-level README with reproduction commands and expected outputs. Checkpoints are included where size permits; otherwise exact training commands and configurations are provided.

## L. C14: Label-Count $\times$ World-Diversity Factorial

A preregistered factorial (design and pre-execution amendment in the artifact package) tests whether the LoRA/full-fine-tuning contrast is governed by the number of super-

vised search states  $N$ , independent of domain and map count. Factors:  $N \in \{256, 1024, 4096, 16384, 65536\}$  exact-count state samples  $\times$  two diversity levels at fixed  $N$  (concentrated: the minimal world count  $w_{\min}(N)$ ; distributed:  $8 \times w_{\min}(N)$  worlds)  $\times$  two domains (static C9 substrate and dynamic C9B substrate, maze-dense target, frozen sources)  $\times$  {unbounded rank-8 LoRA, full fine-tuning, scratch}  $\times$  3 seeds — 180 adaptations, every cell at an identical 2,560 optimizer steps, evaluated on the frozen C9/C9B test cohorts at binding budgets. The measured  $w_{\min}$  spans quantify the domains’ supervision densities: reaching  $N=65,536$  takes 356 static worlds but 4 dynamic worlds ( $\sim 185$  vs.  $\sim 18,900$  usable states per world).

**Preregistered verdict: H-C14 rejected as stated.** On matched-solved expansion ratios the hypothesized crossover point is degenerate: full fine-tuning is at or below LoRA at every tested  $N$  in all 12 (domain  $\times$  diversity  $\times$  seed) cells, and the preregistered regression’s full-FT  $\times$   $\log N$  interaction is null ( $-0.0025$  [ $-0.0087, +0.0028$ ]). State count improves ratios ( $\log_2 N$  coefficient  $-0.0140$  [ $-0.0190, -0.0081$ ]) but identically for both methods.

**The preregistered diversity readout localizes the real effect.** The low-supervision protection LoRA provides appears in *success*, and it is governed by world diversity at fixed  $N$ , not by state count. Concentrated low-world cells collapse full fine-tuning and scratch in both domains (success vs. the frozen zero-shot source:  $-0.300$  [ $-0.550, -0.050$ ] in all three seeds at dynamic concentrated  $N=256$  and  $N=1,024$ ;  $-0.167$  to  $-0.300$  at static concentrated  $N=256$ ), while the distributed cells at the same  $N$  rescue them (dynamic  $+0.150$  to  $+0.200$ ; static  $\approx 0.000$ ; paired dist–conc full-FT deltas  $+0.30$  to  $+0.48$  dynamic,  $+0.233$  static at  $N=256$ ). LoRA never collapses in any of the 60 cells. The dynamic concentrated arm recovers exactly where  $w_{\min}$  forces multiple worlds ( $N=65,536$ , 4 worlds:  $+0.100$  in all seeds), tying the recovery to world count at fixed steps. This refines, without contradicting, the main text’s supervision-density account: in the  $K$ -indexed adaptation protocols map count and state count move together, and C14 separates them — the protective variable is diversity (or adapter-restricted capacity), not raw state count.

**Limitations.** Euclid solves 1/20 dense dynamic test maps at the binding budget, so dynamic matched-ratio cells have  $n=1$  and carry no map-level uncertainty; dynamic inference rests on the 20-map paired success deltas. One target family per domain; three adaptation seeds; adaptations ran on cloud L4 hardware from locally collected, hash-recorded datasets.

| Stage    | Protocol essentials                                                                                                                                                                                                                                                                                   |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C1–C4    | 9 easy suites, $n=80/40$ maps, budgets 100–200; pooled bases, TaskLoRA experts, RBF mixtures.                                                                                                                                                                                                         |
| C5       | calibrated hard maze/dense/rooms; 160 train / $\sim 80$ eval maps; budgets 128–168; scalar residual target with cap.                                                                                                                                                                                  |
| C6       | goal-conditioned raster residual field (8 input channels), sampled at PRM nodes; 96/40 maps; budgets 128–168.                                                                                                                                                                                         |
| C7       | 6 suites (3 held-out families); 96/24 maps per suite; binding budgets 140/24/56; scalar+field $\times$ additive+focal matrix; exact-Dijkstra oracle.                                                                                                                                                  |
| C8       | deterministic moving circular obstacles; space–time PRM; aware ( $W=8$ future window) vs. blind ( $W=0$ ) twins; 53 train / 20 eval maps/suite; 1,129,536 scalar samples.                                                                                                                             |
| C9/C9H   | adapt C7 pooled bases to 3 held-out families; $K \in \{0..32\}$ labeled maps; LoRA(r8, bounded/unbounded), full FT, scratch; C9H matches compute (10 epochs, LR $2 \times 10^{-4}$ ); 30 fixed test maps/target.                                                                                      |
| C9B      | same protocol on C8 dynamic sources (aware+blind), 486 adapters, 3 targets, $K \in \{1, 4, 16\}$ .                                                                                                                                                                                                    |
| C10      | 16 rank-8 experts on 2 family axes; zero-label transfer to 3 interior targets; RBF/nearest/uniform weight-merge and prediction-mix composition.                                                                                                                                                       |
| C11      | compositional missions (waypoint sequences; key/door), configs A/B/C, mission length $K \in \{0, 2, 4, 8\}$ ; 40 train / 25 test maps per cell; 6 learned arms $\times$ 3 seeds (198 checkpoints; 54,450 eval rows); exact mission oracle and leg-sum baseline; forced-compute and would-halt probes. |
| C12      | (A) hidden slow/fast regime dynamics, memory-headroom gate, matched temporal-hierarchy pilot; (B) tied vs. untied product-graph refiner, cycles 1–8, configs A/C $\times$ $K \in \{2, 8\}$ , 3 seeds, 48 checkpoints, two 3,200-row test tables.                                                      |
| C13      | current-state heuristic under bounded observation (radius 0.20 local subgraph); behavior-derived fresh-start rollout labels (median of 3); staged falsification ladder (A–L) then frozen 144-map confirmation (M); architecture substitutions (N/O); persistent-state pilot (P).                      |
| Discrete | $20 \times 20$ to $256 \times 256$ grids; Manhattan-baseline A*; forecasting (dynamic obstacles), additive residual transfer (22 suites $\times$ 3 budgets $\times$ 100 episodes), pooled multi-task bases, learned focal ranking at $w \in \{1.0, 1.05, 1.1\}$ .                                     |

Table 1: Protocol essentials by stage.

| Family              | Generator rule (lengths as fractions of side length)                                                                                                                                   |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Maze / dense maze   | Corridor maze of axis-aligned walls; the dense variant narrows gap width to 0.115. Held out: dense.                                                                                    |
| Rooms / large rooms | $2 \times 2$ rooms from one vertical + one horizontal partition wall, one doorway per wall, doorways biased toward opposite edges; large variant doubles side length. Held out: large. |
| Spiral              | Four serpentine vertical walls spanning $x \in [0.22, 0.78]$ , gap side alternating per wall. Trained.                                                                                 |
| Bugtrap             | Solid vertical back wall, two horizontal arms, and a mouth lip closing the center of the opening (U-pocket). Held out.                                                                 |
| Crossing (dyn.)     | Open arena; dynamic timing control.                                                                                                                                                    |

Table 2: Static geometry per family. Base-anchor families (open/clutter/narrow variants) used by the early stages additionally place 3–68 circular obstacles with radii 0.022–0.115; full ranges ship with the generator in the artifact package.

| Suite       | Patr. | Radius | Span | Period | $v_{ag}$ | $t_{max}$ |
|-------------|-------|--------|------|--------|----------|-----------|
| Maze        | 3     | 0.075  | 0.38 | 0.14   | 0.060    | 110       |
| Rooms       | 3     | 0.075  | 0.40 | 0.14   | 0.060    | 110       |
| Spiral      | 3     | 0.075  | 0.38 | 0.14   | 0.060    | 110       |
| Dense maze  | 4     | 0.062  | 0.42 | 0.17   | 0.060    | 140       |
| Crossing    | 4     | 0.055  | 0.50 | 0.30   | 0.060    | 110       |
| Large rooms | 5     | 0.075  | 0.36 | 0.14   | 0.120    | 130       |

Table 3: Dynamic-suite parameters. Each patroller is a circle sweeping a line segment with triangle-wave motion; radius and span are fractions of the side length, period is a fraction of the horizon  $t_{max} dt$  ( $dt=1$  s), and each patroller crosses its span twice per period. Sweep centers carry lateral jitter  $\leq 0.02$  (0.10 on crossing). The planner and collision checker always use these exact trajectories; only the *aware* heuristic input sees the  $W=8$  future-occupancy channels.

| Model          | Configuration                      | Input                 | Parameters |
|----------------|------------------------------------|-----------------------|------------|
| Scalar HRM     | hid. 192, 2 layers, 4 heads, $k=2$ | 62 tok. $\times$ 16   | 2,158,529  |
| Scalar ON-LSTM | hid. 256, 2 layers, chunk 8        | 62 tok. $\times$ 16   | 1,252,481  |
| Field U-Net    | FiLM-conditioned, 8 input channels | $64 \times 64$ raster | 3,702,370  |
| Field HRM      | 8 input channels                   | $64 \times 64$ raster | 4,016,642  |
| Field ON-LSTM  | 8 input channels                   | $64 \times 64$ raster | 1,448,578  |

Table 4: Instantiated architectures with parameter counts computed from the released constructors. Scalar token sequences: one summary token, 12 nearest-obstacle tokens, 48 ray tokens, one goal token. Dynamic variants append  $W$  future-occupancy channels ( $W=8$  aware;  $W=0$  blind). LoRA inserts rank-8 adapters on the HRM (rank-24 on the ON-LSTM, matching its wider hidden layer); the quoted crossover cells are HRM (rank 8). Optimizer for all models: AdamW, learning rate  $2 \times 10^{-4}$  ( $1.5 \times 10^{-4}$  for adapters), weight decay  $10^{-4}$ , batch 256, gradient clip 1.0, smooth-L1 loss on the capped residual (cap 4.0), 10 base epochs (8 adapter epochs) unless a stage states otherwise.

| Cell        | $K$ | $\Delta\text{succ. [CI]}$ | $\Delta\text{ratio [CI]}$ | $n$ |
|-------------|-----|---------------------------|---------------------------|-----|
| dense/HRM   | 1   | -.240 [-.360, -.133]      | +.112 [+0.052, +.171]     | 10  |
| dense/HRM   | 16  | +.027 [-.020, +.100]      | -.149 [-.179, -.117]      | 10  |
| dense/ON-L. | 1   | -.047 [-.147, +.073]      | -.162 [-.223, -.089]      | 10  |
| dense/ON-L. | 16  | +.000 [-.027, +.033]      | -.126 [-.171, -.083]      | 10  |
| bug./HRM    | 1   | -.467 [-.587, -.340]      | +.225 [+0.089, +.352]     | 13  |
| bug./HRM    | 16  | -.020 [-.120, +.080]      | -.145 [-.239, -.043]      | 16  |
| bug./ON-L.  | 1   | -.093 [-.207, +.007]      | +.229 [+0.117, +.348]     | 15  |
| bug./ON-L.  | 16  | +.100 [+0.013, +.207]     | -.052 [-.154, +.033]      | 15  |
| rl./HRM     | 1   | -.600 [-.773, -.407]      | +.290 [+0.207, +.360]     | 11  |
| rl./HRM     | 16  | +.073 [-.027, +.187]      | -.366 [-.489, -.235]      | 11  |
| rl./ON-L.   | 1   | -.293 [-.433, -.160]      | +.013 [-.156, +.167]      | 11  |
| rl./ON-L.   | 16  | -.053 [-.167, +.053]      | -.170 [-.326, -.010]      | 11  |

Table 5: Direct map-paired full-FT – LoRA contrasts (dense = dense maze, bug. = bugtrap, rl. = large rooms; ON-L. = ON-LSTM). Adaptation seeds are averaged within maps; intervals are 10k map-clustered bootstraps over the 30 test maps per target;  $\Delta\text{ratio}$  is computed on the  $n$  triple-matched maps solved by both methods and the anchor, as a difference of expansion ratios relative to the anchor (negative favors full fine-tuning). At  $K=1$  the success contrast favors LoRA in all six cells (four significantly); at  $K=16$  the effort contrast favors full fine-tuning in all six cells (five significantly) at near-parity success.

| Suite       | Euclid succ. | Field-HRM succ. | Ratio [95% CI]       |
|-------------|--------------|-----------------|----------------------|
| Maze        | 0.583        | 1.000           | 0.521 [0.450, 0.627] |
| Maze-dense  | 0.250        | 0.958           | 0.804 [0.707, 0.829] |
| Rooms       | 0.375        | 0.958           | 0.829 [0.775, 0.885] |
| Spiral      | 0.250        | 0.917           | 0.850 [0.742, 0.919] |
| Bugtrap     | 0.458        | 0.750           | 0.714 [0.533, 0.800] |
| Rooms-large | 0.417        | 0.750           | 0.839 [0.646, 1.222] |

Table 6: C7 static suites at binding budgets: success and matched-solved median expansion ratio (field HRM vs. Euclid). Success gains are BH-significant in 4/6 suites (maze, dense, rooms, spiral); bugtrap and rooms-large reach corrected significance only for other learned arms. Scalar-HRM ratios span 0.427–0.822. Matched  $n$  ranges 6–14.

| Suite       | Euclid succ. | Blind U-Net succ. | Ratio [95% CI]       | $n$ |
|-------------|--------------|-------------------|----------------------|-----|
| Crossing    | 0.30         | 0.95              | 0.191 [0.166, 0.878] | 6   |
| Maze        | 0.30         | 1.00              | 0.093 [0.057, 0.137] | 6   |
| Dense maze  | 0.05         | 0.75              | 0.246 (descriptive)  | 1   |
| Rooms       | 0.30         | 1.00              | 0.099 [0.050, 0.214] | 6   |
| Large rooms | 0.75         | 0.95              | 0.228 [0.185, 0.418] | 15  |
| Spiral      | 0.20         | 0.95              | 0.087 [0.061, 0.287] | 4   |

Table 7: C8 fixed-provider result on the *original* 20-map cohort (field U-Net blind; no future-motion input; additive mode; binding budgets; map-level bootstraps). Success deltas are +0.20 to +0.75 with all six paired CIs excluding zero.

| Suite       | Euclid succ. | Blind U-Net succ. | Ratio [95% CI]       | $n$ |
|-------------|--------------|-------------------|----------------------|-----|
| Crossing    | 0.12         | 0.92              | 0.273 [0.109, 0.516] | 6   |
| Maze        | 0.12         | 0.96              | 0.047 [0.021, 0.153] | 6   |
| Dense maze  | 0.06         | 0.70              | 0.216 [0.100, 0.281] | 3   |
| Rooms       | 0.42         | 1.00              | 0.121 [0.106, 0.172] | 21  |
| Large rooms | 0.82         | 1.00              | 0.250 [0.188, 0.306] | 41  |
| Spiral      | 0.16         | 1.00              | 0.092 [0.065, 0.146] | 8   |

Table 8: C8 fixed-provider *replication* on the fresh 50-map-per-suite cohort (generation seed 999999, frozen before any result was observed; map-disjoint from the original cohort by the evaluation seed formula’s within-suite offset bound). Success deltas are +0.18 to +0.84 with all six paired CIs excluding zero. Map-paired aware–blind success deltas on this cohort: crossing +0.080 [+0.020, +0.160], maze +0.020 [0.000, +0.060], dense maze –0.120 [–0.220, –0.040], rooms 0.000, large rooms –0.200 [–0.320, –0.100], spiral –0.040 [–0.100, 0.000]: one suite significantly favors the aware twin, two the blind twin, three neither.

| Suite       | Seed 1234          | Seed 2001           | Seed 2002          |
|-------------|--------------------|---------------------|--------------------|
| Crossing    | +0.80 [0.68, 0.90] | +0.86 [0.74, 0.96]  | +0.88 [0.78, 0.96] |
| Maze        | +0.84 [0.74, 0.94] | +0.86 [0.76, 0.96]  | +0.84 [0.74, 0.94] |
| Dense maze  | +0.64 [0.50, 0.78] | +0.46 [0.32, 0.60]  | +0.56 [0.42, 0.70] |
| Rooms       | +0.58 [0.44, 0.72] | +0.58 [0.44, 0.72]  | +0.58 [0.44, 0.72] |
| Large rooms | +0.18 [0.08, 0.30] | +0.10 [–0.02, 0.22] | +0.16 [0.06, 0.26] |
| Spiral      | +0.84 [0.74, 0.94] | +0.84 [0.74, 0.94]  | +0.84 [0.74, 0.94] |

Table 9: C8 preregistered multi-seed replication: paired success deltas (blind U-Net – Euclid, map-level 95% CIs) for three independent training seeds on the same frozen fresh cohort. 17 of 18 CIs exclude zero; the single crossing interval is seed 2001’s near-ceiling large-rooms cell (Euclid 0.82).

| Suite (arm)         | Succ. E→L | Ratio [95% CI]       | Matched $n$ |
|---------------------|-----------|----------------------|-------------|
| Crossing (U-Net)    | 0.30→1.00 | 0.258 [0.132, 0.769] | 6           |
| Maze (U-Net)        | 0.30→1.00 | 0.064 [0.046, 0.088] | 6           |
| Maze-dense (U-Net)  | 0.05→0.75 | 0.278 (no CI)        | 1           |
| Rooms (U-Net)       | 0.30→1.00 | 0.096 [0.038, 0.191] | 6           |
| Rooms-large (U-Net) | 0.75→0.90 | 0.380 [0.132, 0.591] | 14          |
| Spiral (field HRM)  | 0.20→0.90 | 0.046 [0.038, 0.073] | 4           |

Table 10: C8 *secondary* view: per-suite strongest arms (explicitly post-hoc selection). Corrected success evidence holds on five suites; rooms-large is supported mainly by expansion evidence. Dense-maze has matched  $n=1$  and supports no expansion inference. Blind arms are the per-suite best in five of six suites.

| Gate                | Verdict                                                                                                                                                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| K=0 continuity      | FAIL, diagnosed as false premise: C7 had no MLP arm and did not give every architecture the C11 input; real shallow separation plus collapsed recurrent cells violate the all-tie assumption. Measurement layer verified.   |
| G1 dose-response    | Negative: no structured arm beats MLP at $\geq 2$ mission depths with non-decreasing gap.                                                                                                                                   |
| G2a forced compute  | Negative: HRM-v2 $k=1/2/4/8$ expansion and MAE curves flat in every $K=8$ config.                                                                                                                                           |
| G2b learned halting | Negative, inverted: pooled Spearman $\rho = -0.407$ , permutation $p \approx 5 \times 10^{-4}$ , $n=675$ ; mean would-halt steps 6.79/7.01/5.30 at $K=2/4/8$ .                                                              |
| G3 closure          | Neither preregistered branch: global-input arms separate at shallow $K$ (U-Net/GNN vs. MLP ratios $\approx 0.67$ – $0.87$ ; state MAE 0.151/0.171 vs. 0.251 at $A/K=0$ ), but no arm converts depth into growing advantage. |

Table 11: C11 preregistered gates.

| Rung | Factor changed                         | $\Delta$ exp. [95% CI]  |
|------|----------------------------------------|-------------------------|
| C    | fresh certifier (oracle rank)          | +11.50 (0/6 wins)       |
| D    | shared-queue certifier (oracle rank)   | −6.83 (6/6 wins)        |
| E    | exact rollout rank                     | +1.33 (2/6)             |
| F    | monotone recalibration                 | +1.33                   |
| G    | analytic local escape                  | +6.83 (0/6)             |
| I    | one-family model, live six suites      | +15.36 [+12.02, +18.64] |
| J    | suite-balanced training, static insert | +16.17 [+9.67, +22.67]  |
| K    | + one local Bellman backup             | −1.21 [−8.04, +5.46]    |
| L    | + scale $\alpha=1.50$ (48 fresh maps)  | −13.04 [−18.67, −7.65]  |
| M    | frozen 144-map confirmation            | −12.96 [−16.30, −9.74]  |

Table 12: The C13 falsification ladder. Target repairs (E, F), analytic constructions (G), and distribution repairs (I, J) fail; the integration repair (K) plus scale calibration (L) flip the sign, and the frozen confirmation (M) reproduces it.

| Suite       | Local  | Field HRM | $\Delta$ | 95% CI           | W/T/L    |
|-------------|--------|-----------|----------|------------------|----------|
| Maze        | 47.63  | 84.38     | −36.75   | [−45.58, −27.83] | 24/0/0   |
| Dense maze  | 95.38  | 104.50    | −9.13    | [−15.21, −3.13]  | 18/1/5   |
| Rooms       | 98.04  | 106.17    | −8.13    | [−14.33, −1.71]  | 16/0/8   |
| Spiral      | 120.25 | 121.33    | −1.08    | [−5.88, +4.21]   | 14/1/9   |
| Bugtrap     | 14.96  | 24.29     | −9.33    | [−17.79, −2.92]  | 18/1/5   |
| Large rooms | 33.58  | 46.92     | −13.33   | [−18.92, −7.50]  | 19/0/5   |
| Pooled      | 68.31  | 81.26     | −12.96   | [−16.30, −9.74]  | 109/3/32 |

Table 13: C13-M per-suite paired expansion deltas versus the complete-map field-HRM comparator. Every suite mean is negative; the spiral interval crosses zero, and the preregistered gate is pooled with a suite-breadth condition (at least four negative suite means), which passes. The local provider also beats every other fixed C7 learned provider in pooled expansions on this cohort.

| Cell (HRM)          | $\Delta$ success [CI]   | map ratio [CI]              |
|---------------------|-------------------------|-----------------------------|
| dense LoRA $K=1$    | +0.467 [+0.300, +0.633] | 0.686 [0.590, 0.774]        |
| dense LoRA $K=16$   | +0.433 [+0.267, +0.600] | 0.786 [0.694, 0.891]        |
| dense full $K=1$    | +0.227 [+0.080, +0.380] | 0.789 [0.751, 0.836]        |
| dense full $K=16$   | +0.460 [+0.287, +0.633] | <b>0.610</b> [0.549, 0.670] |
| dense scratch $K=1$ | −0.167 [−0.300, −0.047] | 1.035 [1.029, 1.039]        |
| rl. full $K=1$      | −0.167 [−0.340, +0.013] | 1.293 [1.170, 1.503]        |
| rl. full $K=8$      | +0.387 [+0.227, +0.547] | 0.590 [0.498, 0.712]        |
| bug. LoRA $K=32$    | −0.320 [−0.507, −0.133] | 1.153 [1.012, 1.523]        |

Table 14: Map-clustered C9 key cells (binding budgets, additive mode; dense = maze-dense, rl. = rooms-large, bug. = bugtrap). The  $K=16$  full-FT/LoRA CIs are disjoint (the crossover), the rooms-large  $K=1$  catastrophe excludes parity, and the bugtrap LoRA degradation is significant in both metrics.

| Suite       | $w_h$ | WA*  | Blind | $\Delta$ succ. [CI] | Bl./WA* (n) | Sub. W/B    |
|-------------|-------|------|-------|---------------------|-------------|-------------|
| Crossing    | 1.5   | 1.00 | 0.92  | −.08 [−.16, −.02]   | 0.825 (46)  | 1.008/1.164 |
| Maze        | 5     | 0.96 | 0.96  | +0.00 [−.06, +.06]  | 0.601 (47)  | 1.017/1.012 |
| Dense maze  | 5     | 0.70 | 0.70  | +0.00 [−.06, +.06]  | 0.982 (34)  | 1.006/1.008 |
| Rooms       | 3     | 0.98 | 1.00  | +0.02 [+0.00, +.06] | 0.554 (49)  | 1.015/1.019 |
| Large rooms | 1.5   | 0.98 | 1.00  | +0.02 [+0.00, +.06] | 0.728 (49)  | 1.003/1.082 |
| Spiral      | 5     | 0.70 | 1.00  | +0.30 [+0.18, +.42] | 0.217 (35)  | 1.013/1.011 |

Table 15: Tuned weighted A\* versus the fixed blind U-Net on the confirmation cohort at binding budgets.  $w_h$  is the per-suite inflation selected on the development cohort under the pre-registered rule (highest success, ties toward smaller  $w_h$ ).  $\Delta$ succ. is the paired blind-weighted success difference (10k map bootstraps); Blind/WA\* is the median expansion ratio on the  $n$  jointly solved maps (below 1 = learned expands fewer nodes); sub. is mean empirical suboptimality (arrival over optimal arrival) on solved maps. The learned heuristic expands fewer nodes in every suite, wins spiral success outright, and loses crossing success to inflation; results reported as-is per the frozen design.